

# Machine Learning Task 1 - Sales & Demand Forecasting

## MACHINE LEARNING TASK 1 Sales & Demand Forecasting for Businesses

**Objective:**  
Build a demand forecasting model using historical sales data to predict future sales trends.

**Dataset:**  
Store Sales - Time Series Forecasting (Kaggle)  
Daily sales data from 2022-01-01 to 2024-12-31

**Model Used:**  
Linear Regression (Time-based forecasting)

**Forecast Horizon:**  
Next 30 Days



★ **Goal:** Help businesses make smarter decisions with accurate sales forecasts.

### 2. Monthly Sales Distribution



### 4. Python Code (Sample)

```
# Feature Engineering
df['Day'] = df['Date'].dt.day
df['Month'] = df['Date'].dt.month
df['Tenth'] = range(len(df))

# Model Training
X = df[['Day', 'Month', 'Trend']]
y = df['Sales']
model = LinearRegression()
model.fit(X, y)

# Forecast Next 30 Days
future_pred = model.predict(future_df[['Day', 'Month', 'Trend']])
```

### Key Takeaways

- Upward sales trend with strong weekly seasonality observed.
- Linear Regression captured the overall trend effectively.
- Forecast can support better business decision-making.
- Advanced models can further improve forecast accuracy.



### Dataset Summary

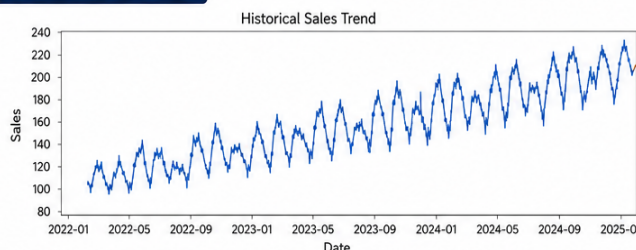
- Source:** Store Sales - Time Series Forecasting (Kaggle)
- Period:** 2022-01-01 to 2024-12-31
- Frequency:** Daily
- Total Records:** 1,095 days
- Features Used:** Date, Sales



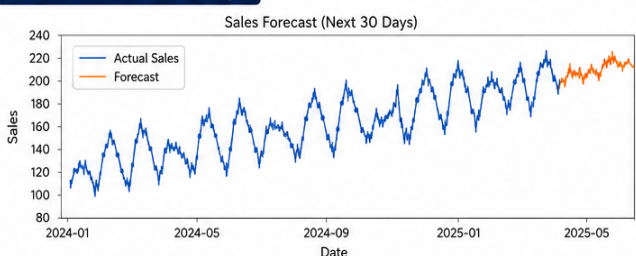
### Tools & Libraries Used

- Python
- Pandas, NumPy
- Scikit-learn
- Matplotlib

### 1. Historical Sales Trend



### 3. Sales Forecast (Next 30 Days)



### 5. Model Evaluation

| Metric | Value |
|--------|-------|
| MAE    | 12.87 |
| RMSE   | 14.32 |
| MAPE   | 8.21% |



The model performs well and captures the overall sales trend with low errors.

### 6. Business Impact



This forecast helps businesses in:

- Inventory Planning
- Staffing & Workforce Management
- Cash Flow Planning
- Sales & Marketing Strategy

By predicting upcoming demand, businesses can reduce overstocking and avoid stockouts.